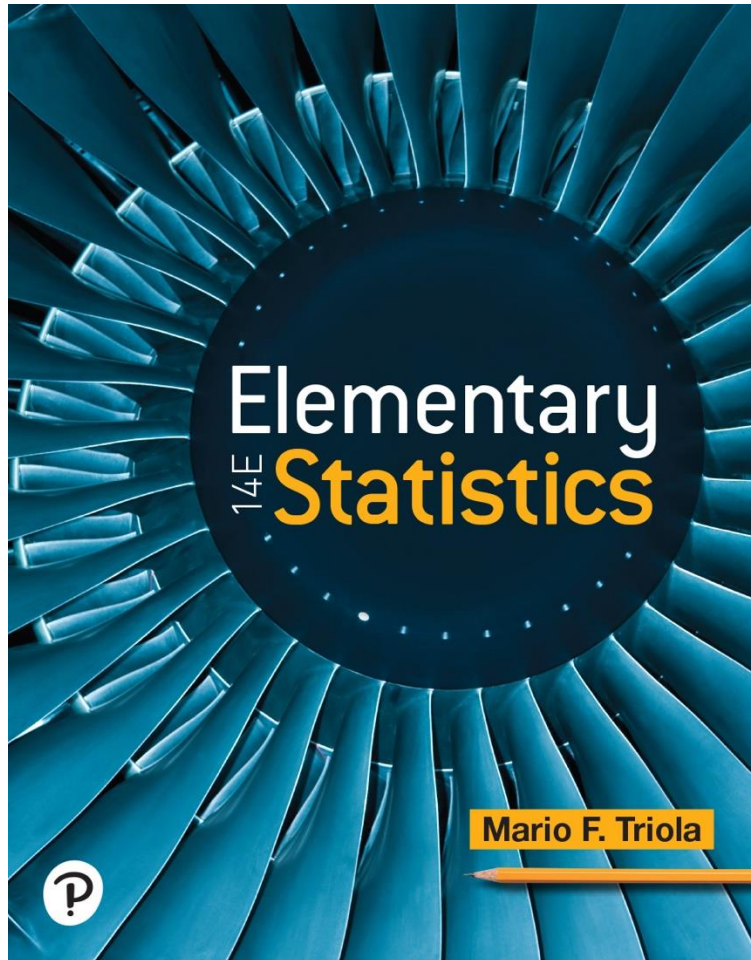


Elementary Statistics

Fourteenth Edition



Chapter 4 Probability

Probability

4-1 Basic Concepts of Probability

4-2 Addition Rule and Multiplication Rule

4-3 Complements and Conditional Probability

4-4 Counting

Key Concept

This section requires much greater use of formulas as we consider five different methods for counting the number of possible outcomes in a variety of situations. Not all counting problems can be solved with these five methods, but they do provide a strong foundation for the most common real applications.

Multiplication Counting Rule

1. Multiplication Counting Rule

- For a sequence of events in which the first event can occur n_1 ways, the second event can occur n_2 ways, the third event can occur n_3 ways, and so on, the total number of outcomes is $n_1 \cdot n_2 \cdot n_3 \dots$

Example: Multiplication Counting Rule: Hacker Guessing a Passcode (1 of 2)

A computer hacker finds that a password is entered as •••••, so the characters are hidden, but we can see that there are five characters. We can use 92 different characters with a typical keyboard. How many different passwords are possible using five characters? If the hacker starts to generate all different possibilities, what is the probability of guessing the correct password on the first attempt?

Example: Multiplication Counting Rule; Hacker Guessing a Passcode (2 of 2)

Solution

There are 92 different possibilities for each character, so the total number of different possible passwords is

$$\begin{aligned}n_1 \cdot n_2 \cdot n_3 \cdot n_4 \cdot n_5 &= 92 \cdot 92 \cdot 92 \cdot 92 \cdot 92 \\ &= 6,590,815,232.\end{aligned}$$

Because all of the passwords are equally likely, the probability of getting the correct password on the first attempt is $1/6,590,815,232$ or 0.000000000152.

Unfortunately, there are sophisticated programs that allow hackers to try large numbers of likely possibilities, beginning with the commonly used but highly ineffective password of “password.”

Factorial Rule (1 of 2)

2. Factorial Rule

- The number of different **arrangements** (order matters) of n different items when all n of them are selected is **$n!$** .

Factorial Rule (2 of 2)

Notation

The **factorial symbol (!)** denotes the product of decreasing positive whole numbers. For example, $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$. By special definition, $0! = 1$.

The factorial rule is based on the principle that the first item may be selected n different ways, the second item may be selected $n - 1$ ways, and so on. This rule is really the multiplication counting rule modified for the elimination of one item on each selection.

Example: Factorial Rule: Scrambling Letters (1 of 3)

Some words consist of letters which can be rearranged to form other words. Consider the word “steam.”

- a. How many different ways can the letters of “steam” be arranged?
- b. If the letters of “steam” are arranged randomly, what is the probability that the letters will be in alphabetical order?
- c. How many different words can be formed with the letters in “steam”?

Example: Factorial Rule: Scrambling Letters (2 of 3)

Solution

a. For those 5 different letters, the number of different arrangements is $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$.

Note that this solution could have been done by applying the multiplication counting rule. The first letter can be any one of the 5 letters in "steam", the second letter can be any one of the 4 remaining letters, and so on. The result is again $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$.

Example: Factorial Rule: Scrambling Letters (3 of 3)

Solution

b. There is only one arrangement with the letters listed in alphabetical order, aemst, so the probability is $1/120$.

c. The letters of “steam” can be arranged to form these other words: teams, meats, mates, and tames, so there are five different words that can be formed. This is found by trial and error, and statistics is no help here.

Permutations and Combinations:

Does Order Count? (1 of 2)

- Permutations
 - **Permutations** of items are arrangements in which different sequences of the same items are counted **separately**. (The letter arrangements of abc, acb, bac, bca, cab, and cba are all counted **separately** as six different permutations.)

Permutations and Combinations:

Does Order Count? (2 of 2)

- Combinations
 - **Combinations** of items are arrangements in which different sequences of the same items are counted as being the **same**. (The letter arrangements of abc, acb, bac, bca, cab, and cba are all considered to be the **same** combination.)

Permutations Rule (1 of 2)

3. Permutations Rule (When All of the Items Are Different)

- When n different items are available and r of them are selected without replacement, the number of different permutations (order counts) is given by

$${}_nP_r = \frac{n!}{(n - r)!}$$

Example: Permutations Rule (with Different Items): Trifecta Bet (1 of 4)

In a horse race, a **trifecta** bet is won by correctly selecting the horses that finish first and second and third, and you must select them in the correct order. The 144th running of the Kentucky Derby had a field of 20 horses.

Example: Permutations Rule (with Different Items): Trifecta Bet (2 of 4)

- a. How many different trifecta bets are possible?
- b. If a bettor randomly selects three of those horses for a trifecta bet, what is the probability of winning by selecting Justify to win, Good Magic to finish second, and Audible to finish third, as they did? Do all of the different possible trifecta bets have the same chance of winning? (Ignore “dead heats” in which horses tie for a win.)

Example: Permutations Rule (with Different Items): Trifecta Bet (3 of 4)

Solution

a. There are $n = 20$ horses available, and we must select $r = 3$ of them without replacement. The number of different sequences of arrangements is found as shown:

$${}_nP_r = \frac{n!}{(n-r)!} = \frac{20!}{(20-3)!} = 6840$$

Example: Permutations Rule (with Different Items): Trifecta Bet (4 of 4)

Solution

b. There are 6840 different possible arrangements of 3 horses selected from the 20 that are available. If one of those arrangements is randomly selected, there is a probability of $1/6840$ that the winning arrangement is selected.

There are 6840 different possible trifecta bets, but not all of them have the same chance of winning, because some horses tend to be faster than others.

Permutations Rule (2 of 2)

4. Permutations Rule (When Some Items Are Identical to Others)

The number of different permutations (order counts) when n items are available and all n of them are selected **without replacement**, but some of the items are identical to others, is found as follows:

$$\frac{n!}{n_1! n_2! \dots n_k!}$$

where n_1 are alike, n_2 are alike, . . . , and n_k are alike.

Example: Permutations Rule (with Some Identical Items): Designing Surveys (1 of 3)

When designing surveys, pollsters often repeat a question to see if a subject is thoughtlessly providing answers just to finish quickly. For one survey with 10 questions, 2 of the questions are identical to each other (with superficial differences in wording), and 3 other questions are also identical to each other (with superficial differences in wording). For this survey, how many different arrangements are possible? Is it practical to survey enough subjects so that every different possible arrangement is used?

Example: Permutations Rule (with Some Identical Items): Designing Surveys (2 of 3)

Solution

We have 10 questions with 2 that are identical to each other and 3 others that are also identical to each other, and we want the number of permutations. Using the rule for permutations with some items identical to others, we get

$$\frac{n!}{n_1! n_2! \dots n_k!} = \frac{10!}{2! 3!} = \frac{3,628,800}{2 \cdot 6} = 302,400$$

Example: Permutations Rule (with Some Identical Items): Designing Surveys (3 of 3)

Interpretation

There are 302,400 different possible arrangements of the 10 questions. It is not practical to accommodate every possible permutation. For typical surveys, the number of respondents is somewhere around 1000.

Combinations Rule

5. Combinations Rule:

When n different items are available, but only r of them are selected **without replacement**, the number of different combinations (order does not matter) is found as follows:

$${}_nC_r = \frac{n!}{(n-r)!r!}$$

Example: Combinations Rule:

Lottery (1 of 3)

In Florida's Cash 5 lottery game, winning the jackpot requires that you select 5 different numbers from 1 to 35, and the same 5 numbers must be drawn in the lottery. The winning numbers can be drawn in any order, so order does not make a difference.

- a. How many different lottery tickets are possible?
- b. Find the probability of winning the jackpot when one ticket is purchased.

Example: Combinations Rule:

Lottery (2 of 3)

Solution

a. There are $n = 35$ different numbers available, and we must select $r = 5$ of them without replacement (because the selected numbers must be different). Because order does not count, we need to find the number of different possible **combinations**. We get

$${}_nC_r = \frac{n!}{(n-r)!r!} = \frac{35!}{(35-5)!5!} = \frac{35!}{30! \cdot 5!} = 324,632$$

Example: Combinations Rule:

Lottery (3 of 3)

Solution

b. If you select one 5-number combination, your probability of winning is $1/324,632$.